

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 03



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Recap of Previous Lecture



Topic

Terminologies related to graphs



Topic

Different types of graphs



Topics to be Covered



Topic

Bipartite graph



Topic

Complete bipartite graph



Topic

Sum of degree theorem



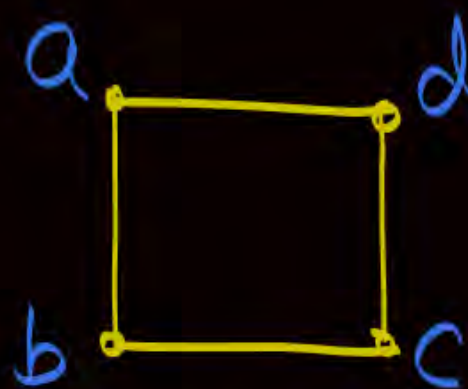
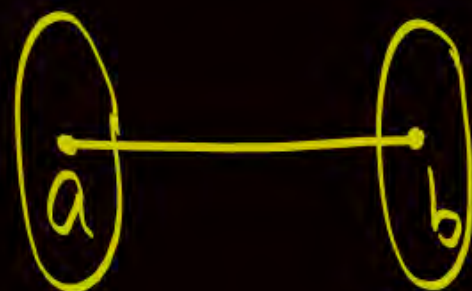
Topic

Degree Sequence

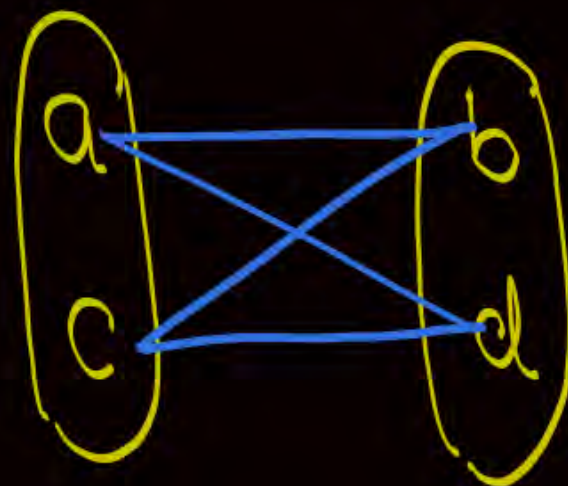




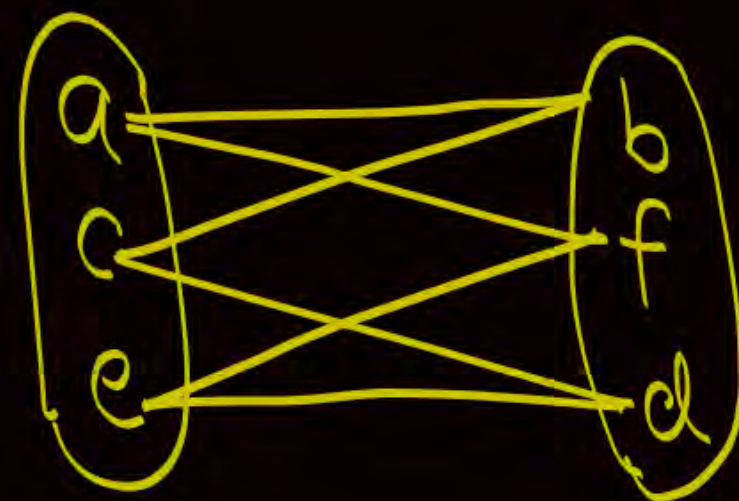
G_1



G_2



G_3



All are bipartite graphs

Note:-

① In a graph if all cycles are of even length.
then that graph is always a bipartite graph

② Tree is an acyclic graph { i.e. cycles of length = 0 }
∴ Every tree is a bipartite graph

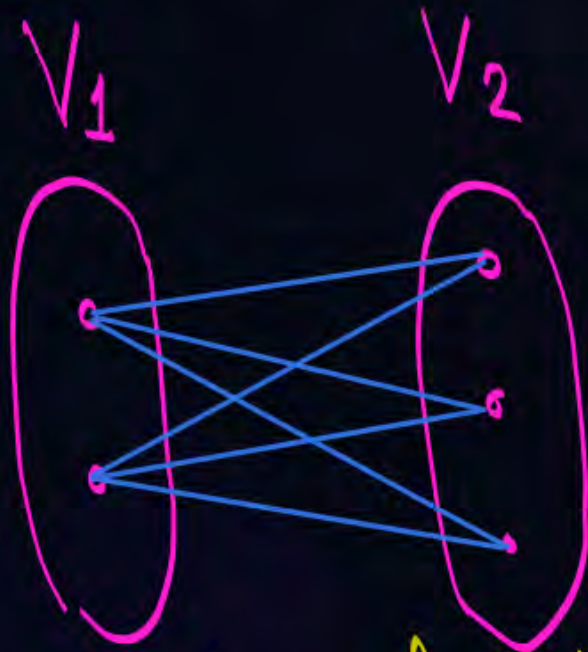
↑
Even

③ If there exist any cycle of odd length,
then graph can not be a bipartite graph



Topic : Complete bipartite Graph

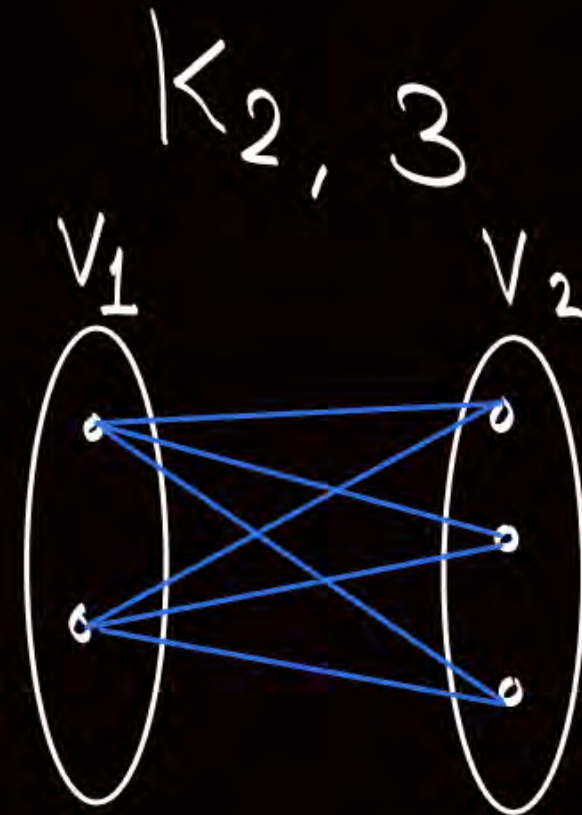
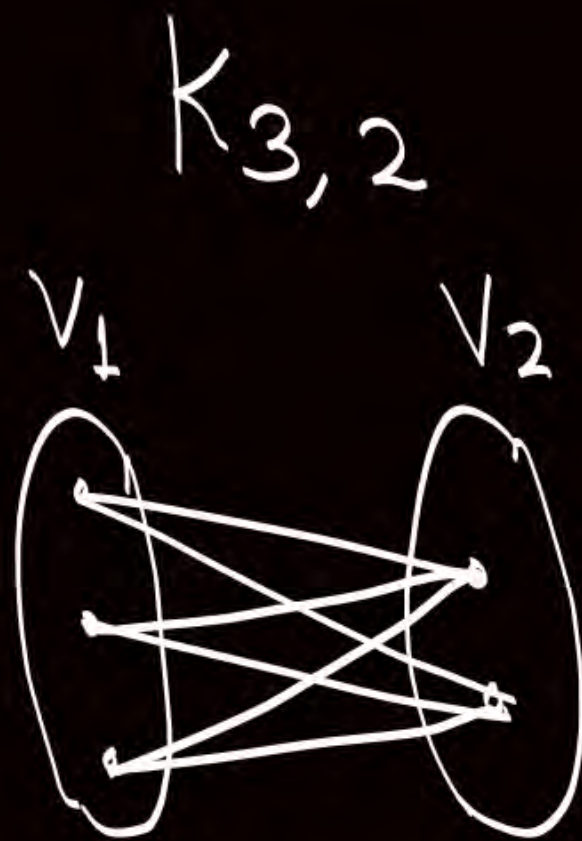
A bipartite graph $G=(V_1, V_2, E)$ is called a complete bipartite graph if every vertex in set V_1 is adjacent to every vertex of set V_2 .



→ Every Complete bipartite graph is also a bipartite graph, but Every bipartite graph need not be a Complete bipartite graph

Every vertex of set V_1 is adjacent to every vertex of set V_2
∴ It is a Complete bipartite graph

* A complete bipartite graph $G = (V_1, V_2, E)$
Where $|V_1| = m$ and $|V_2| = n$ is
represented by $K_{m,n}$



→ In a Complete bipartite graph $K_{m,n}$

$$|E(K_{m,n})| = m \cdot n$$

In general, Complete bipartite graph is not
a Complete graph

A complete bipartite graph $K_{m,n}$
is a Complete graph if and only if $m=n=1$



$$K_{1,1} \equiv K_2$$

Q:- Let G be a bipartite graph with ' n ' vertices,
 then maximum number of edges possible in graph G is?
 $|V| = n$ (let $n = \text{Even}$)

$ V_1 $	$ V_2 $	max $ E $
1	$(n-1)$	$1 \cdot (n-1)$
2	$(n-2)$	$2(n-2)$
3	$(n-3)$	$3 \cdot (n-3)$
$\vdots$	$\vdots$	$\vdots$
$n/2$	$n/2$	$\frac{n}{2} \cdot \frac{n}{2} = \frac{n^2}{4}$
$\vdots$	$\vdots$	$\vdots$

$$|V| = 10$$

$ V_1 $	$ V_2 $	$ E $
1	9	9
2	8	16
3	7	21
4	6	24
5	5	25
6	4	24
7	3	21
8	2	16
9	1	9

$$= \frac{n}{2} * \frac{n}{2} = \frac{n^2}{4}$$

$$|V| = 9$$

$|V_1|$

$|V_2|$

$|E|$

1

8

8

2

7

14

3

6

18

4

5

20

5

4

20

6

3

18

7

2

14

8

1

8

$$= \frac{n-1}{2} * \frac{n+1}{2}$$

$$= \frac{n+1}{2} * \frac{n-1}{2}$$

Q:- Let G be a bipartite graph with ' n ' vertices, then maximum number of edges possible in graph G is?

$$\text{Maximum no. of edges possible} = \begin{cases} \frac{n^2}{4} & \text{if } n = \text{Even} \\ \left(\frac{n-1}{2} \times \frac{n+1}{2}\right) & \text{if } n = \text{odd} \end{cases}$$

$$\text{Maximum no. of edges possible} = \left\lfloor \frac{n^2}{4} \right\rfloor$$

K-partite graph:-

If we can partition the set of vertices into 'K' subsets such that no two vertices of same subset are adjacent to each other, then it is a K-partite graph.

→ If graph is a K -partite graph then graph is $(K+1)$ partite as well { Provided $(K+1)$ is less than or equal to the total number of vertices in the graph }

→ If graph is a K -partite graph, then it need not be $(K-1)$ partite

H.W.

Q: A forest with n -vertices and k components
will have how many edges?

i.e., 'k' trees



# Vertices	# trees in forest (i.e. no. of components)	# Edges
n	1	$(n-1)$
n	2	$(n-1)-1 = (n-2)$
n	3	$(n-2)-1 = (n-3)$
$\vdots$	$\vdots$	$\vdots$
n	k	$(n-k)$



Topic : Sum of degree theorem

Handshaking lemma

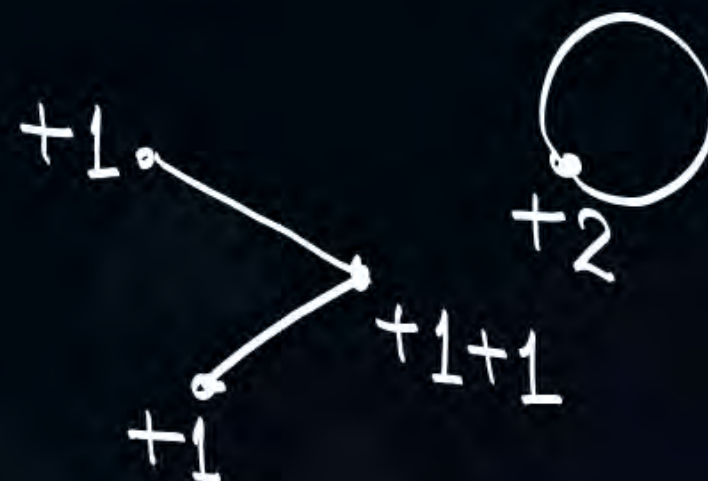


Let $G=(V,E)$ be a graph

Where $V = \{v_1, v_2, v_3, \dots, v_{|V|}\}$

then

$$\sum_{i=1}^{|V|} \deg(v_i) = 2|E|$$



$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

Corollary 1:-

If graph $G = (V, E)$ is a directed graph,

then

$$\sum_{i=1}^{|V|} \deg^+(v_i) = \sum_{i=1}^{|V|} \deg^-(v_i) = |E|$$

Corollary 2:-

We know

$$\sum_{i=1}^{|V|} \deg(V_i) = \underbrace{2|E|}$$

it is always even

∴ In a graph 'G'.
Number of vertices
of Odd degree
are always Even

Corollary 3:

In a graph G if degree of Each vertex is exactly ' k ' i.e. graph is k -regular

then

$$k \cdot |V| = 2|E|$$

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$\underbrace{k + k + \dots + k}_{|V| \text{ times } k} = 2|E|$$

$$\therefore k|V| = 2|E|$$

Corollary 4

In a graph G if degree of each vertex is at least k { i.e. $\delta(G) \geq k$ }

then

$$k \cdot |V| \leq 2|E|$$

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$(k + x_1) + (k + x_2) + \dots + (k + x_{|V|}) = 2|E|$$

$$x_1 \geq 0$$

$$k|V| + \sum_{i=1}^{|V|} x_i = 2|E|$$

— this value is ≥ 0

Corollary 5:-

In a graph G if degree of each vertex is at most K { i.e. $\Delta(G) \leq K$ }

then

$$K \cdot |V| \geq 2|E|$$

$$\deg(v_1) + \deg(v_2) + \dots + \deg(v_{|V|}) = 2|E|$$

$$(K - x_1) + (K - x_2) + \dots + (K - x_{|V|}) = 2|E|$$

$$x_1 \geq 0$$

$$K|V| - \sum_{i=1}^{|V|} x_i = 2|E|$$

— this value is ≥ 0

Note:-

In a graph G ,

degree is at least $\delta(G)$

and degree is at most $\Delta(G)$

$\therefore$

$$\delta(G) \cdot |V| \leq 2|E| \leq \Delta(G) \cdot |V|$$

#Q. The number of edges in a k - regular graph with n vertices is

$$k|V| = 2|E|$$

$$k \cdot n = 2|E|$$

$$|E| = \frac{nk}{2}$$

#Q. A non-directed graph contains 16 edges and all vertices are of degree 2. Then the number of vertices in G is _____.

$$k|V| = 2|E|$$

$$2 * |V| = 2 * 16$$

$$\boxed{|V| = 16}$$

#Q. G is undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

$$\hookrightarrow K|V| \leq 2|E|$$

$$3 * n \leq 2 * 25$$

$$n \leq \frac{50}{3}$$

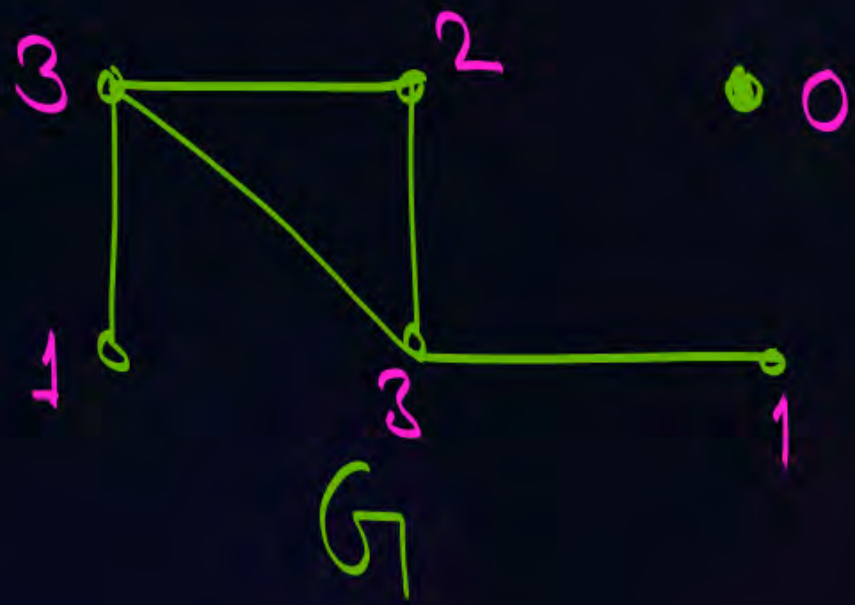
$$n \leq 16.66$$

$$\boxed{n_{\max} = 16}$$



Topic : Degree sequence

In a graph G if degrees of all the vertices are arranged in non-increasing or non-decreasing order, then it is called degree sequence of graph G .



degree sequence of G is
 $\{3, 3, 2, 1, 1, 0\}$

Graphic:- If there exists a simple non-directed graph w.r.t given degree sequence then that degree sequence is called a graphic

#Q. Which of the following degree sequences represent a simple non-directed graph? { i.e. Which of the following is/are graphic? }

1. $\{2, 3, 3, 4, 4, 5\}$

2. $\{2, 3, 4, 4, 5\}$

3. $\{1, 3, 3, 3\}$

4. $\{0, 1, 2, 3, \dots, n-1\}$

5. $\{1, 3, 3, 4, 5, 6, 6\}$

6. $\{3, 3, 3, 3, 2\}$

No. of vertices with odd degree can never be odd in any graph { be it simple or not }
 i. Given sequence does not represent any graph

#Q. Which of the following degree sequences represent a simple non-directed graph? {i.e. Which of the following is/are graphic?}

~~1.~~ {2, 3, 3, 4, 4, 5}

~~2.~~ {2, 3, 4, 4, 5} $\Rightarrow$ '5' degrees in given sequence $\therefore |V|=5$

3. {1, 3, 3, 3}

4. {0, 1, 2, 3, ..., n-1}

5. {1, 3, 3, 4, 5, 6, 6}

6. {3, 3, 3, 3, 2}

In a simple graph with n -vertices
degree of each vertex is $\leq (n-1)$

$\therefore$ With '5' vertices, a vertex of
deg=5 is not possible in a simple graph

#Q. Which of the following degree sequences represent a simple non-directed graph? {i.e. Which of the following is/are graphic?}

~~1.~~ {2, 3, 3, 4, 4, 5}

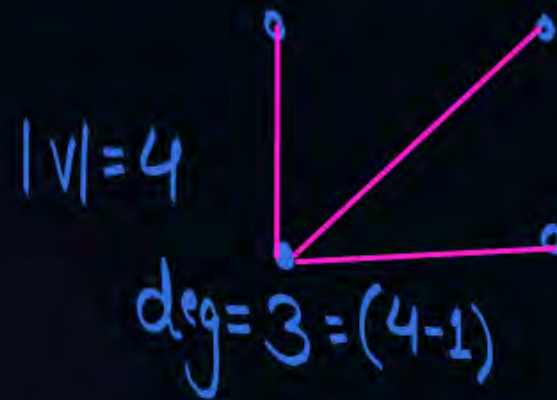
~~2.~~ {2, 3, 4, 4, 5}

3. {1, 3, 3, 3}

4. {0, 1, 2, 3, ..., n-1}

5. {1, 3, 3, 4, 5, 6, 6}

6. {3, 3, 3, 3, 2}



In a simple graph with 'n' vertices, if there is '1' vertex of degree $= (n-1)$, then degree of every other vertex will be at least '1'



In a simple graph if there are two vertices of degree $= (n-1)$, then degree of every other vertex will be at least '2'

#Q. Which of the following degree sequences represent a simple non-directed graph? { i.e. Which of the following is/are graphic? }

~~1.~~ {2, 3, 3, 4, 4, 5}

~~2.~~ {2, 3, 4, 4, 5}

~~3.~~ {1, 3, 3, 3}

4. {0, 1, 2, 3, ..., n-1}

5. {1, 3, 3, 4, 5, 6, 6}

6. {3, 3, 3, 3, 2}

In a simple graph with n -vertices if there are ' k ' vertices of degree $= (n-1)$, then degree of every other vertex will be at least ' k '

With ' 4 ' vertices if there are 3 vertices of $\text{deg} = (4-1) = 3$, then deg of every other vertex must be at least ' 3 ',
 $\therefore$ Vertex of $\text{deg} = 1$ is not possible

#Q. Which of the following degree sequences represent a simple non-directed graph? { i.e. Which of the following is/are graphic? }

1. {2, 3, 3, 4, 4, 5}

2. {2, 3, 4, 4, 5}

3. {1, 3, 3, 3}

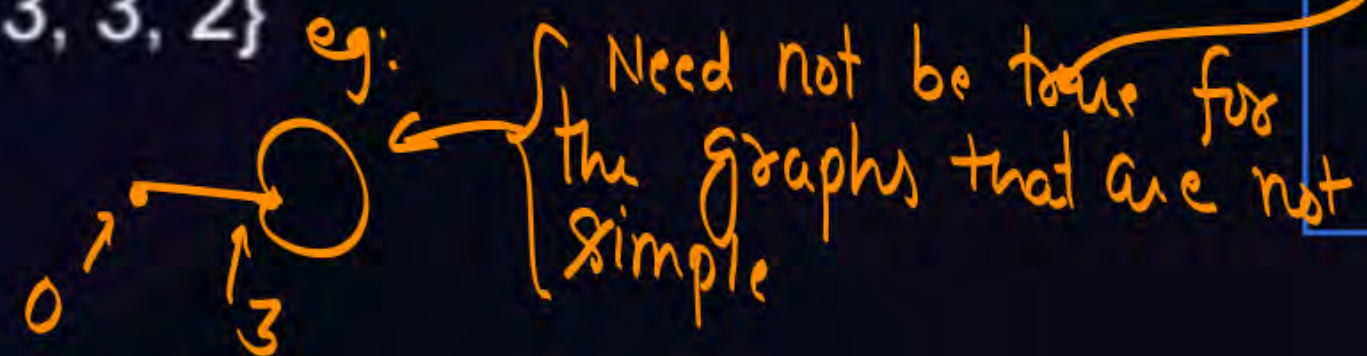
4. {0, 1, 2, 3, ..., n-1} $\Rightarrow$ 0 to (n-1) we have 'n' vertices

5. {1, 3, 3, 4, 5, 6, 6}

6. {3, 3, 3, 3, 2}

With 'n' vertices there is one vertex of $\text{deg} = 'n-1'$, $\therefore$ deg of Every other vertex must be at least '1'
 $\therefore$ vertex of $\text{deg} = 0$ is not possible in a simple graph

Note: In a simple graph all vertices can never have distinct degrees. i.e., in a simple graph there must be at least '2' vertices with same degree



#Q. Which of the following degree sequences represent a simple non-directed graph? { i.e. Which of the following is/are graphic? }

~~1.~~ {2, 3, 3, 4, 4, 5}

~~2.~~ {2, 3, 4, 4, 5}

~~3.~~ {1, 3, 3, 3}

~~4.~~ {0, 1, 2, 3, ..., n-1}

~~5.~~ {1, 3, 3, 4, 5, 6, 6} = $|V| = 7 = n$

6. {3, 3, 3, 3, 2}

Two vertices of $\deg = (n-1)$

∴ vertex of $\deg = 1$ is not possible in a simple graph

#Q. Which of the following degree sequences represent a simple non-directed graph? {i.e. Which of the following is/are graphic?}

1. $\{2, 3, 3, 4, 4, 5\}$
2. $\{2, 3, 4, 4, 5\}$
3. $\{1, 3, 3, 3\}$
4. $\{0, 1, 2, 3, \dots, n-1\}$
5. $\{1, 3, 3, 4, 5, 6, 6\}$
6. $\{3, 3, 3, 3, 2\}$



i.e. $\{3, 3, 3, 3, 2\}$ is a graphic

H.W.

Read about 'Havel Hakimi's' algo.



2 mins Summary



✓ **Topic**

Bipartite graph

✓ **Topic**

Complete bipartite graph

✓ **Topic**

Sum of degree theorem

✓ **Topic**

Degree Sequence

THANK - YOU